

Micro Energy Harvesting 2026

Exercise 3: Thermoelectric converters

Task 1: TEGs - Thermoelectric Generators

a) How many thermocouples are needed to obtain a desired output voltage of 2.5 V, for example to operate a microprocessor, from a temperature difference of $\Delta T = 10\text{ K}$ by $20\text{ }^\circ\text{C}$? Determine the required number of unicouples for the following material combinations:

- Nichrome (NiCr alloy with additions of Mn, Fe, Si, and C) & Constantan (CuNi alloy)
- n- Bi_2Te_3 & p- $(\text{Bi,Sb})_2\text{Te}_3$

b) A typical TEG, such as the Thermalforce TEG 049-150-30 has a $25 \times 25\text{ mm}^2$ footprint and consists of 49 thermocouples of Bi_2Te_3 & $(\text{Bi,Sb})_2\text{Te}_3$. How large would the footprint and edge length of a similar generator with 257 thermocouples be?

c) Using the data sheet, you can identify the values for the thermal resistance (or conductivity) of the generator at 300 K, as well as the Seebeck voltage $\alpha_{\text{generator}}$. Use these values to calculate ZT @ 293 K.

d) A typical thermal conductivity value for Bi_2Te_3 is $\kappa = 2\text{ W/(K}\cdot\text{m)}$. Use this value to determine the ratio of leg-length to footprint $L_{\text{Leg}}/A_{\text{Leg}}$ of a single thermo leg. Using this value, you can additionally calculate the ratio of active area A_{act} to total area A_{Gen} of this TEG. The length of the legs can be calculated from the data sheet.

Task 2: μ TEGs

Using the data sheet for the Micropelt "MPG-D751" μ -thermogenerator, calculate ZT @ 293 K and L/A ratios for the thermo legs and the ratio of active area to generator area as before. The specific electrical resistance can be assumed with the typical value of $\rho = 10^{-5}\text{ }\Omega\text{m}$. The area of the thermo legs can be taken from the data sheet. Are these values significantly different from the macro generator? How high (long) are the thermolegs and why are they not made longer?

Task 3: Thermoelectric Power Generation

A typical DC-DC converter such as the one from Linear Technologies (bipolar, LTC3109, $U_{\text{start}} = 27\text{ mV}$) is the only load connected to a TEG and has an internal resistance of $3\text{ }\Omega$. Calculate the output power (at the load resistance) for a generator with 144 thermocouples (Seebeck coefficient: $400\text{ }\mu\text{V/K}$, internal resistance $R_g = 10\text{ }\Omega$, thermal resistance $K_g = 6.1\text{ K/W}$), which has the heat sink of **Task 4** ($K_{\text{HS}} = 4\text{ K/W}$) coupled to its hot side to generate a temperature difference between heat sink and cold side of the generator of 15 K . Also calculate the output voltage at the generator. At which temperature difference between the heat sink and the cold side does the DC-DC converter start to run?

Task 4: TEG Optimization

A heat sink from Fischer electronic (ICK S 25 x 25 x 18) is to be used in a busy road tunnel together with an **ideal TEG**. By constantly passing cars and trucks, forced convection occurs on average, so that the heat sink has a thermal resistance of 4 K/W.

a) The generator is a standard macroscopic TEG, with a leg length of $L=1.7\text{ mm}$ and the leg area $A=1.65\text{ mm}^2$. How many thermocouples are needed so that the thermogenerator has the same thermal resistance as the heat sink (neglect contact resistances, material Bi_2Te_3)?

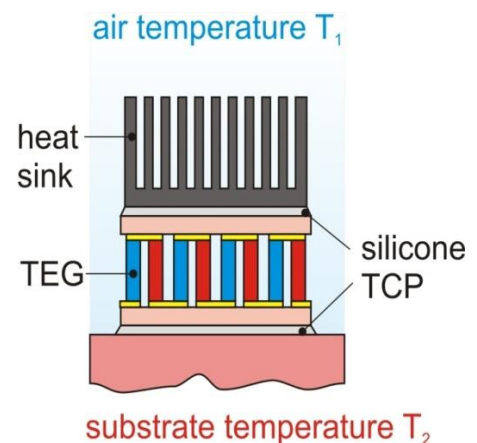
b) What is the ratio of active to total area?

c) Adjust the area A of the legs, so that 144 thermocouples fit on the generator (25 % active area), how long would the thermo legs have to be to adjust the generator thermally ideal and what is their edge length?

d) What is the internal resistance of this generator?

Task 5: Thermoelectric Harvesting System

The following model of a wall-mounted system is given. The Fischer Elektronik ICK S 25 x 25 x 18 heat sink (Task 4) is used together with Micropelt D651 micro generators to harvest from the temperature difference between warm air and cold wall ($T_{\text{cold}} = 27.5\text{ }^\circ\text{C}$). As the heat sink is much larger than the individual generator, it is decided to equip the heat sink with as many generators of this type as possible. The air speed of 9 km/h on this hot day is due to a radial fan set up, which blows the warm air ($T_{\text{hot}} = 35.5\text{ }^\circ\text{C}$) through the fins. The contact resistances due to the thermal paste between heat sink and generator or generator and wall are $K_{\text{TCP}} = 0.05\text{ K/W}$.



Calculate the output voltage of the generators without load when operated in parallel? How high is it when used serially? Also calculate the output power for both cases for a load resistance of $R_L = 100\ \Omega$. Any comments on the practicality of this system?